

UnMute

An Assistive Web-Based Platform for Sign Language Learning & Communication

Towards Easier Communication and a More Inclusive World

PROJECT OVERVIEW & CHALLENGE

About UnMute

UnMute is an assistive web-based platform designed to support deaf, hard-of-hearing, and mute users. It provides digital tools that foster communication, sign language learning, social interaction, and seamless access to supportive resources.

The Problem

Individuals relying on sign language frequently encounter communication barriers in daily life, limited access to tailored educational resources, and social isolation. UnMute mitigates these challenges through intelligent digital solutions.

CORE PLATFORM CAPABILITIES

Translator Workspace

- Visual sign representations for text inputs.
- Real-time speech-to-text input to facilitate dialogue.
- AI prototype for hand gesture recognition via camera.

Sign Language Learning

- Structured educational content for sign language practice.
- Interactive exercises tailored for skill development.
- Progress tracking and module comprehension.

Social Community Platform

A dedicated, safe digital space facilitating peer communication, social interaction, knowledge sharing, and community engagement among platform users.

SUPPORTING TOOLS & ACCESSIBILITY

Daily Tasks Management

Integrated task organization and tracking tools designed to help users structure daily activities and boost independence.

Interactive Location Map

Map-based exploration tool to help users locate supportive community centers, accessible services, and relevant places.

Accessibility & Platform Administration

- **Dual Language Support:** Full Arabic and English user interface accessibility.
- **User Customization:** Tailored accessibility options for personalized visual and navigation experiences.
- **Admin Control Panel:** Comprehensive dashboard for managing platform content, users, and analytical insights.

TECHNOLOGY ARCHITECTURE (TECH STACK)

The system architecture integrates modern web technologies with advanced Computer Vision frameworks:

PHP & MySQL

JavaScript ES6

HTML5 & CSS3

MediaPipe

OpenCV

Leaflet API

OpenStreetMap

Web Speech API

EXPECTED IMPACT & SOCIAL VALUE

UnMute aims to foster independence, streamline communication, improve access to learning, and encourage sustainable social inclusion through an intuitive, accessible, and user-friendly digital environment.

UnMute — Communication is for Everyone.